

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.metrics import mean_absolute_error
```

```
df = pd.read_csv("/content/placement.csv")
df.head()
```

	cgpa	package
0	6.89	3.26
1	5.12	1.98
2	7.82	3.25
3	7.42	3.67
4	6.94	3.57

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.isnull().sum()
```

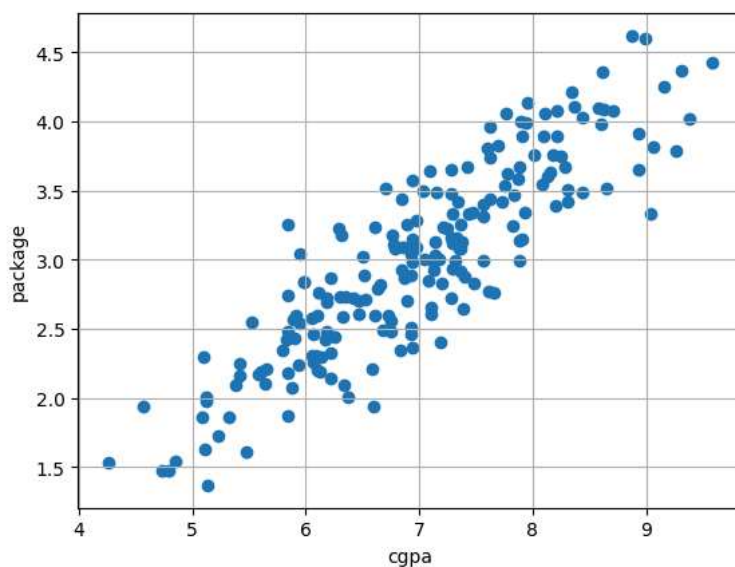
```

      0
cgpa  0
package 0
dtype: int64
```

```
df.duplicated().sum()
```

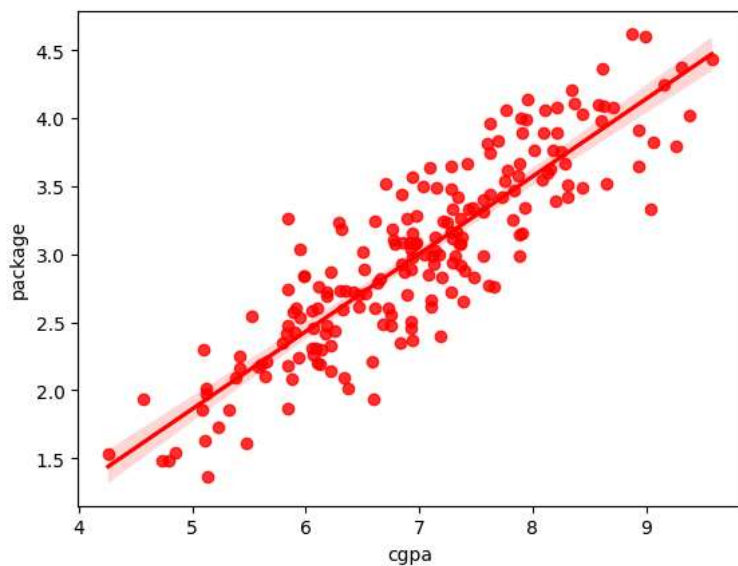
```
np.int64(0)
```

```
plt.scatter(df["cgpa"],df["package"])
plt.xlabel("cgpa")
plt.ylabel("package")
plt.grid(True)
plt.show()
```



```
sns.regplot(x="cgpa", y="package", data=df,color = "r")
```

```
plt.show()
```



```
x = df.iloc[:,0:1]
y = df.iloc[:, -1]
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=3)
```

```
x_train.shape
```

```
(160, 1)
```

```
x_test.shape
```

```
(40, 1)
```

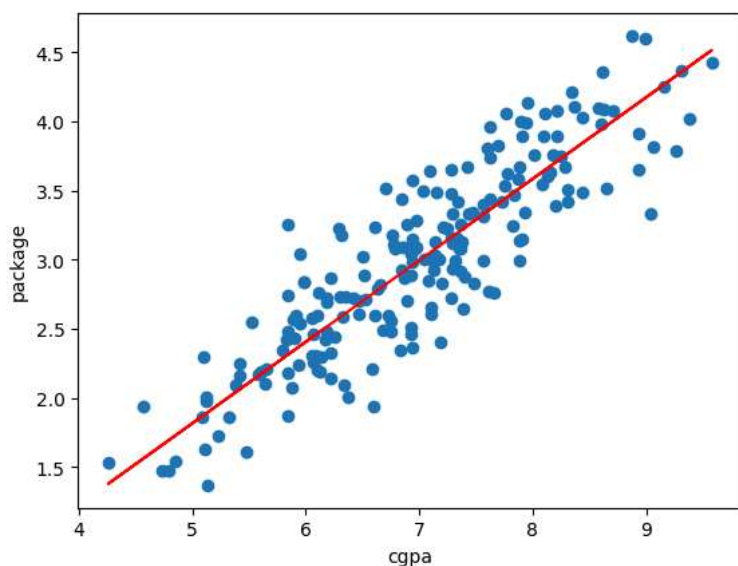
```
lr = LinearRegression()
```

```
lr.fit(x_train,y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
plt.scatter(df["cgpa"],df["package"])
plt.plot(x_train,lr.predict(x_train),color = "r")
plt.xlabel("cgpa")
plt.ylabel("package")
plt.show()
```



```
y_pred = lr.predict(x_test)
y_pred
```

```
array([4.39751239, 4.32680345, 3.10707437, 3.11885919, 2.5355105 ,
       2.1583962 , 3.16010607, 2.44712434, 2.28213683, 3.55489761,
       3.07761232, 3.84362575, 3.16010607, 2.51783327, 4.00272085,
       2.84780829, 2.77709936, 3.24259983, 2.93030204, 2.94797928,
       3.42526457, 1.7282502 , 4.1382463 , 2.83602347, 2.33516853,
       3.17189089, 3.54311279, 1.88734529, 2.82423864, 3.35455564,
       3.20724536, 2.31159889, 2.37641541, 3.19546054, 3.94968915,
       2.34106094, 2.9597641 , 3.45472662, 2.70639042, 2.31159889])
```

```
np.array(y_test)
```

```
array([4.02, 3.79, 2.4 , 3.24, 2.87, 2.17, 2.72, 2.26, 2.35, 4.14, 3.03,
       3.49, 3.65, 2.69, 4.08, 2.56, 2.79, 3.67, 2.7 , 3.09, 3.42, 1.54,
       3.65, 2.6 , 2.08, 3.33, 3.34, 2.01, 3.52, 2.77, 2.92, 3.26, 3.04,
       3.42, 4.36, 2.57, 3.57, 3.62, 2.89, 2.18])
```

```
y_test.values
```

```
array([4.02, 3.79, 2.4 , 3.24, 2.87, 2.17, 2.72, 2.26, 2.35, 4.14, 3.03,
       3.49, 3.65, 2.69, 4.08, 2.56, 2.79, 3.67, 2.7 , 3.09, 3.42, 1.54,
       3.65, 2.6 , 2.08, 3.33, 3.34, 2.01, 3.52, 2.77, 2.92, 3.26, 3.04,
       3.42, 4.36, 2.57, 3.57, 3.62, 2.89, 2.18])
```

```
print("mean_absolute_error", "=", mean_absolute_error(y_test, y_pred))
```

```
mean_absolute_error = 0.31001787170563705
```

```
print("mean_squared_error", "=", mean_squared_error(y_test, y_pred))
```

```
mean_squared_error = 0.14551953158290803
```

```
print("RMSE", "=", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
RMSE = 0.38147022371727524
```

```
r2 = r2_score(y_test, y_pred)
print("r2_score", "=", r2)
```

```
r2_score = 0.651675487510094
```

```
adjusted_r2 = 1 - ((1 - r2) * (40 - 1) / (40 - 1 - 1))
print("Adjusted R2 Score =", adjusted_r2)
```

```
Adjusted R2 Score = 0.6425090529708859
```

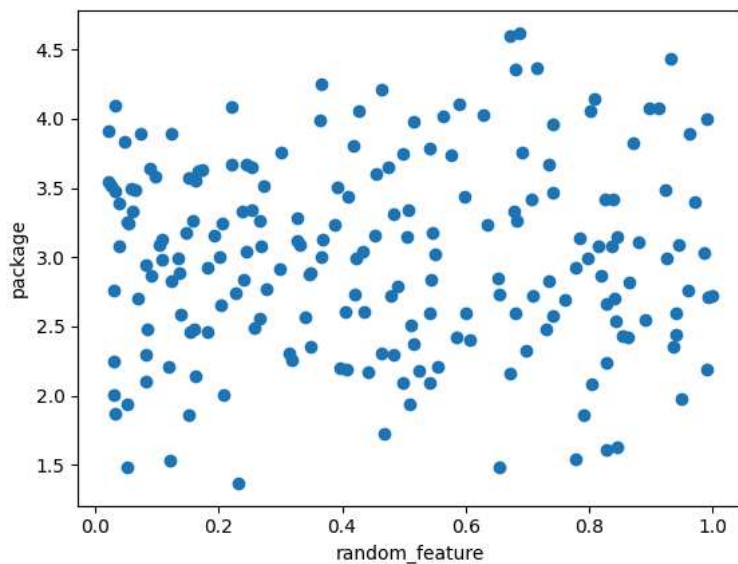
```
new_df = df.copy()
new_df["random_feature"] = np.random.random(200)
new_df = new_df[["cgpa", "random_feature", "package"]]
new_df.head()
```

	cgpa	random_feature	package
0	6.89	0.265828	3.26
1	5.12	0.949033	1.98
2	7.82	0.052706	3.25
3	7.42	0.244630	3.67
4	6.94	0.152095	3.57

Next steps:

[Generate code with new_df](#)
[New interactive sheet](#)

```
plt.scatter(new_df["random_feature"], new_df["package"])
plt.xlabel("random_feature")
plt.ylabel("package")
plt.show()
```



```
X = new_df.iloc[:,0:2]
Y = new_df.iloc[:, -1]
```

```
X_train,X_test,Y_train,Y_test = train_test_split(X,Y,test_size=0.2,random_state=3)
```

```
X_train.shape,X_test.shape
```

```
((160, 2), (40, 2))
```

X_train

	cgpa	random_feature	
156	7.38	0.108911	
115	8.35	0.463783	
155	6.98	0.104528	
15	7.25	0.387044	
61	5.48	0.829326	
...	...	...	
0	6.89	0.265828	
184	7.88	0.784262	
131	6.37	0.208344	
152	7.37	0.836405	
106	6.13	0.990992	

160 rows × 2 columns

Next steps: [Generate code with X_train](#) [New interactive sheet](#)

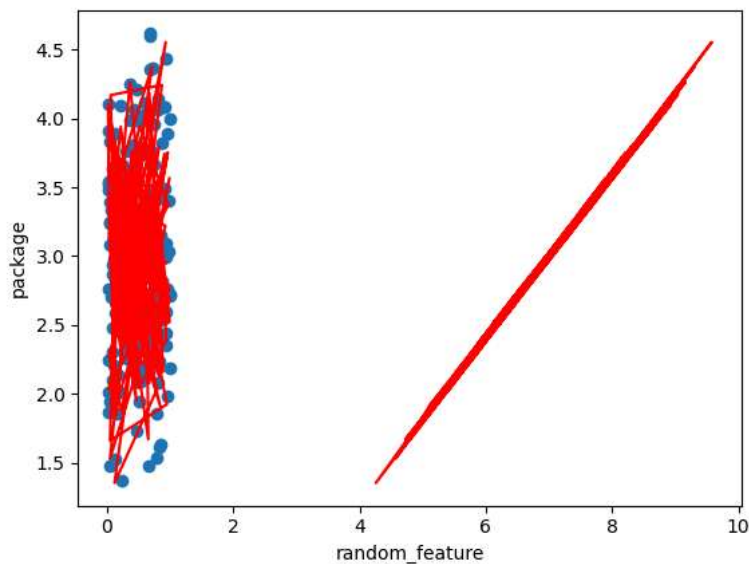
```
lr1 = LinearRegression()
```

```
lr1.fit(X_train,Y_train)
```

LinearRegression ⓘ ?

```
LinearRegression()
```

```
plt.scatter(new_df["random_feature"],new_df["package"])
plt.plot(X_train,lr1.predict(X_train),color = "r")
plt.xlabel("random_feature")
plt.ylabel("package")
plt.show()
```



```
y_pred1 = lr1.predict(X_test)
y_pred1
```

```
array([4.40697576, 4.33443904, 3.11867452, 3.08854995, 2.50756311,
        2.15669569, 3.17945928, 2.43639124, 2.31809444, 3.58222177,
        3.11786054, 3.87989676, 3.16166023, 2.54066039, 4.0371636 ,
        2.83337622, 2.77957838, 3.22688122, 2.95944673, 2.93856047,
        3.45476547, 1.75162404, 4.12393511, 2.84698032, 2.36100989,
        3.15561854, 3.54760218, 1.85417906, 2.81032517, 3.34133469,
        3.19569423, 2.32822445, 2.37431462, 3.21463802, 3.96755429,
        2.33182596, 2.93680472, 3.43322355, 2.68199952, 2.31631861])
```

```
np.array(Y_test)
```

```
array([4.02, 3.79, 2.4 , 3.24, 2.87, 2.17, 2.72, 2.26, 2.35, 4.14, 3.03,
        3.49, 3.65, 2.69, 4.08, 2.56, 2.79, 3.67, 2.7 , 3.09, 3.42, 1.54,
        3.65, 2.6 , 2.08, 3.33, 3.34, 2.01, 3.52, 2.77, 2.92, 3.26, 3.04,
        3.42, 4.36, 2.57, 3.57, 3.62, 2.89, 2.18])
```

```
print("mean_absolute_error", "=", mean_absolute_error(Y_test, y_pred1))
```

```
mean_absolute_error = 0.3159861869685788
```

```
print("mean_squared_error", "=", mean_squared_error(Y_test, y_pred1))
```

```
mean_squared_error = 0.14829086950848647
```

```
print("RMSE", "=", np.sqrt(mean_squared_error(Y_test, y_pred1)))
```

```
RMSE = 0.3850855353145408
```

```
R2 = r2_score(Y_test, y_pred1)
print("r2_score", "=", R2)
```

```
r2_score = 0.6450418423810076
```

```
X_test.shape
```

```
(40, 2)
```

```
#adjusted_r2 = 1 - ((1 - r2) * (n - 1) / (n - p - 1)) X_test.shape where n = X_test.shape[0] and p = X_test.shape[1]
```

```
adjusted_R2 = 1 - ((1 - R2) * (40 - 1) / (40 - 2 - 1))
print("Adjusted R2 Score =", adjusted_R2)
```

```
Adjusted R2 Score = 0.6258549149421431
```

